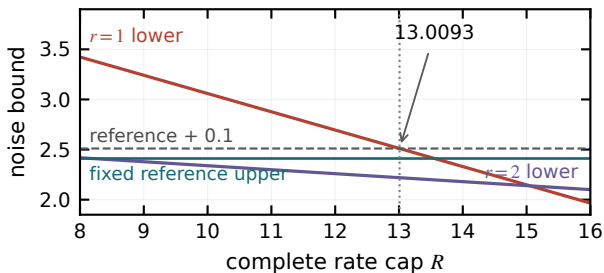
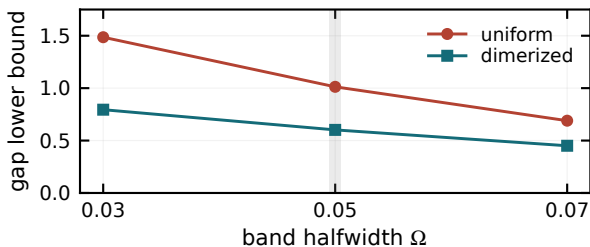


(a) Necessary rate, uniform chain



(b) Bandwidth, every $N \geq 64$



(c) Minimum range, $R=8$

